

BERLIN DAHLEM, Germany

WMO: 103810

Lat: 52.4536N

Lon: 13.3017E

Elev: 262

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.4	15.6	3.7	6.7	12.8	8.9	8.8	18.1	20.2	42.7	17.4	41.4	5.8	110	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	84.7	66.2	81.1	64.7	78.1	63.3	68.4	79.8	66.7	77.3	65.0	74.5	7.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	92.3	72.1	62.9	86.5	70.0	61.2	81.4	69.3	32.8	79.5	31.4	77.7	30.1	74.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.5	13.1	DB	9.7	91.7	8.5	3.6	94.3	-1.3	96.4	-6.1	98.4	-12.3	101.0		
(5)			WB	8.9	71.9	8.2	1.6	73.1	-1.8	74.0	-6.4	74.9	-12.4	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	33.1	32.8	40.6	47.5	56.8	61.4	65.6	64.7	57.4	49.5	40.1	35.5
	DBStd	13.61	9.76	8.58	6.62	6.71	6.99	6.38	5.94	5.96	4.93	6.56	6.17	6.96
	HDD50	2286	525	482	295	125	19	1	0	0	2	90	298	449
	HDD65	6102	990	901	755	524	265	146	66	81	233	482	746	913
	CDD50	1871	0	1	5	51	231	342	484	456	225	73	2	1
	CDD65	213	0	0	0	0	13	37	85	72	6	0	0	0
	CDH74	1763	0	0	0	5	107	351	730	527	41	2	0	0
	CDH80	422	0	0	0	0	6	83	189	142	2	0	0	0
Wind	WSAvg	6.7	7.6	7.1	7.4	6.7	6.3	6.6	6.0	5.5	6.0	6.4	6.9	7.3
Precipitation	PrecAvg	22.7	1.8	1.3	1.5	1.4	2.1	2.6	2.5	2.4	1.8	1.6	1.9	1.9
	PrecMax	32.9	4.7	3.5	4.6	3.5	5.8	7.6	7.3	6.3	5.3	5.7	4.0	5.1
	PrecMin	14.8	0.2	0.1	0.4	0.0	0.2	0.1	0.3	0.3	0.2	0.2	0.0	0.4
	PrecStd	4.3	0.9	0.8	0.9	0.8	1.2	1.7	1.4	1.3	1.0	1.1	0.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	57.3	62.9	72.9	80.5	87.5	89.2	89.8	78.8	70.6	55.7	53.4
		MCWB	48.0	49.5	50.9	55.5	61.9	66.4	68.2	68.6	64.8	60.5	51.4	49.2
	2%	DB	48.5	51.0	56.8	67.5	77.0	81.8	84.9	83.5	74.1	65.5	53.2	50.9
		MCWB	45.3	46.3	48.2	53.7	59.9	64.3	66.3	66.7	62.2	57.3	49.8	47.7
	5%	DB	46.6	46.8	53.0	63.2	73.8	78.0	81.5	79.8	69.9	61.5	50.9	48.7
		MCWB	43.6	43.1	45.9	51.3	58.3	62.4	64.9	65.1	60.2	55.3	47.9	46.3
	10%	DB	44.5	43.4	50.2	59.4	70.1	73.9	78.3	75.8	66.2	58.8	48.7	46.0
		MCWB	41.6	40.0	45.0	49.5	56.6	60.0	64.0	63.6	58.1	53.4	46.1	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	50.5	52.5	56.8	65.0	69.1	71.2	70.7	66.2	62.0	52.7	49.9
		MCDB	51.3	55.9	60.3	70.0	75.8	81.7	83.3	85.5	75.6	66.8	54.6	51.8
	2%	WB	46.2	46.6	49.4	54.4	62.2	66.1	68.7	68.2	63.8	58.6	50.3	48.4
		MCDB	48.3	50.8	54.5	65.0	73.7	78.4	80.0	79.4	70.7	63.8	52.6	50.6
	5%	WB	44.0	43.4	47.3	52.4	59.6	63.8	66.7	66.7	61.8	56.2	48.2	46.2
		MCDB	46.3	46.6	51.9	61.2	69.8	73.8	77.8	76.4	67.7	60.6	50.2	48.4
	10%	WB	41.8	40.3	45.2	50.2	57.7	61.6	64.8	64.9	59.8	54.3	46.5	43.8
		MCDB	44.1	42.9	49.9	58.2	67.9	71.2	75.0	73.4	65.3	58.0	48.4	46.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	10.0	12.1	15.1	17.3	16.3	17.2	17.3	14.8	13.0	8.3	7.0
		MCDBR	8.5	14.1	17.4	21.4	22.8	23.4	23.6	22.8	20.6	17.0	9.9	8.7
		MCWBR	7.2	9.8	10.2	10.3	9.8	9.4	9.2	9.5	10.2	9.9	7.9	7.6
	5% WB	MCDBR	8.5	13.9	16.1	19.3	19.9	20.8	20.8	20.4	18.0	15.4	9.2	8.8
		MCWBR	7.7	10.2	10.1	9.6	9.8	9.7	10.1	9.4	10.0	9.8	8.0	8.0
Clear-Sky Solar Irradiance	taub	0.326	0.352	0.393	0.417	0.414	0.411	0.432	0.425	0.388	0.375	0.350	0.320	
	taud	2.310	2.285	2.192	2.201	2.252	2.285	2.238	2.270	2.352	2.346	2.350	2.322	
	Ebn at Noon	207	236	248	258	266	267	259	253	250	225	196	187	
	Edh at Noon	20	27	36	41	41	40	41	38	31	26	20	17	
All-Sky Solar Radiation	RadAvg	200	422	746	1248	1544	1643	1539	1324	966	536	246	149	
	RadStd	22	73	83	161	184	130	204	106	110	64	31	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page